UNIVERSITY OF LIVERPOOL | English Language Centre

COMP208: Assessment Support

Session 2: Requirements Analysis (CA1)

Session aim

The main aim of this session is to help you to write the Requirements Analysis (CA1) assignment effectively. We will:

- analyse examples from previous student projects
- identify features of effective academic and technical writing
- examine common weaknesses using tutor feedback
- evaluate and improve sentences to communicate project ideas clearly

Task 1: Reviewing the required sections

1. **The instructions state that your Requirements Analysis must include the following sections (within a 5-page limit). Read through the section list and consider:**
   - What is the purpose of each section, what should it include, and what does it communicate to the reader?
   - Which sections are likely to require the most page space in your own project?

1) Title
2) Project Description
3) Aims and Objectives
4) Key Literature and Background Reading
5) Development and Implementation Summary
6) Data Sources
7) Testing and Evaluation
8) Ethical Considerations
9) BCS Project Criteria
10) UI / UX Mockup
11) Project Plan (including Risks and Contingency Plans)
12) References

Suggested answers below

1) Title

Identifies the project and gives the reader a sense of what the system is about.

2) Project Description

Briefly explains what the software will do and the problem it aims to address.

3) Aims and Objectives

States the overall purpose of the project (aims) and the specific, achievable outcomes (objectives).

4) Key Literature and Background Reading

Shows understanding of existing research, software, or technologies and justifies project decisions.

5) Development and Implementation Summary

Outlines how the system will be built, including components, technologies, and development approach.

6) Data Sources

Explains what data the system will use, where it comes from, and how it will be handled responsibly.

7) Testing and Evaluation

Describes how the system will be tested and how its success will be evaluated.

8) Ethical Considerations

Identifies ethical issues specific to the project and how guidelines will be followed.

9) BCS Project Criteria

Explains how the project meets the BCS requirements using real project examples.

10) UI / UX Mockup

Shows what the system will look like and explains how users will interact with it.

11) Project Plan (including Risks and Contingency Plans)

Outlines the project timeline, dependencies, risks, and how potential problems will be managed.

12) References

Lists all sources cited in the document using an appropriate academic referencing style.

Task 2: Aims and Objectives

1. **Read the definitions of aims and objectives in the box below. Then read examples A–H (taken from a group project) and decide whether each one is an aim or an objective.**

Aims	Objectives
The broad purpose of the project, what you want to achieve overall. You do not need a lot of aims. Tip: If it can be tested or measured, it's probably not an aim.	Specific actions or outcomes the project will deliver to achieve the aims. There are usually more objectives than aims. Tip: A good objective can be tested or demonstrated later.

A User accounts and secure encryption of passwords.

B Provide an accessible and flexible alternative to existing spreadsheet tools.

C Filtering, sorting and searching of cell content using natural language or data types.

D Support collaborative work on shared spreadsheets between multiple users.

E Client-side formulae capture and performance of basic arithmetic.

F Develop a web-based spreadsheet application for creating, managing, and analysing tabular data.

G Dynamic grid layout of labelled rows and columns with simple grid resizing functionality.

H Real-time collaboration and sharing of documents with other users with a permissions system.

2. **What do you notice about the wording of this group's objectives? Are there any other ways (or styles) you could use to write the objectives?**

Suggested answers below

A User accounts and secure encryption of passwords.
Objective

B Provide an accessible and flexible alternative to existing spreadsheet tools.
Aim

C Filtering, sorting and searching of cell content using natural language or data types.
Objective

D Support collaborative work on shared spreadsheets between multiple users.
Aim

E Client-side formulae capture and performance of basic arithmetic.
Objective

F Develop a web-based spreadsheet application for creating, managing, and analysing tabular data.
Aim

G Dynamic grid layout of labelled rows and columns with simple grid resizing functionality.
Objective

H Real-time collaboration and sharing of documents with other users with a permissions system.
Objective

2. What do you notice about the wording of this group's objectives? Are there any other ways (or styles) you could use to write the objectives?

- The objectives are written as short noun phrases, not full sentences.
- They focus on specific, testable features and system functionality.
- Many read like a technical checklist rather than outcomes.

Other possible styles:

- Writing objectives as full sentences using action verbs (e.g. implement, design, enable).
- Framing objectives as outcomes for the user or system rather than features.
- Explicitly linking objectives to what will be tested or demonstrated later.

3. **Below are three weak objectives. Identify what the problem is with each.**
   - **A** Ensure that all system responses are delivered within 2 seconds, with a success rate of at least 98%.
   - **B** Implement the backend system using a suitable framework and integrate all required APIs.
   - **C** Provide an advanced, innovative, and user-friendly platform for all users.

Suggested answers below

Suggested answers below

A Ensure that all system responses are delivered within 2 seconds, with a success rate of at least 98%.

- This focuses on non-functional performance targets rather than what the system will do.
- It is overly specific and reads more like a testing requirement than an objective.

B Implement the backend system using a suitable framework and integrate all required APIs.

- This is too vague ("suitable framework", "required APIs").
- It describes a development task, not a clear, testable outcome of the project.

C Provide an advanced, innovative, and user-friendly platform for all users.

- This is too generic and subjective.
- Terms like advanced, innovative, and user-friendly are not defined or measurable.

Task 3: Literature and Background Reading

1. **Read the example below, taken from a project about a delivery route planning and optimisation app, and discuss the following questions:**
   1. What types of sources have the group used in this section?
   2. How does the writer use these sources to justify features or decisions in their own project?
   3. What phrases or sentence structures show that the sources are being used to support the project, rather than just described?

One useful feature of Upper [1] is that it allows users to upload routes from a spreadsheet. We would take a similar approach by allowing users to add routes to a database, which would then be used to display routes to drivers within the app. A further example is Routific [2], which provides a graphics-focused UI using a variety of graphs and visual elements to present useful information. For example, the number of completed and skipped stops is shown using a pie chart, which makes progress and delivery data easier to interpret. Both apps provide route updates based on live traffic information, which is a function we would want to implement to ensure the route is always optimal. Research has shown that incorporating real-time traffic information into vehicle routing can improve routing efficiency and decision-making in delivery systems [3]. To help inform the Ethical Guidelines, we reviewed industry guidance on data privacy and security in mobile applications [4], ethical considerations surrounding the use of AI in app development [5], and broader ethical responsibilities for mobile app developers [6].

[1] Upper Inc. (2025). *Delivery Route Planning & Route Optimization Software*. [online]. Available at: https://www.upperinc.com/

[2] Routific.com (2024). *Delivery Route Planning Software & Route Optimization Software*. [online]. Available at: https://www.routific.com/

[3] Kim, S., Lewis, M. E., and White III, C. C. (2005). *Optimal Vehicle Routing With Real-Time Traffic Information*. *IEEE Transactions on Intelligent Transportation Systems*.

[4] MoldStud (no date). *The Ethics of Mobile App Development: Privacy and Security*. [online]. Available at: https://moldstud.com/articles/p-the-ethics-of-mobile-app-development-privacy-and-security (Accessed: 13 February 2025).

[5] SayOne Technologies (no date). *AI in Mobile App Development: Ethical Considerations*. [online]. Available at: https://www.sayonetech.com/blog/ai-mobile-app-development/ (Accessed: 13 February 2025).

[6] ESP Workforce (no date). *Ethical Considerations for Mobile App Developers*. [online]. Available at: https://www.espworkforce.com/blog/ethical-considerations-for-mobile-app-developers/ (Accessed: 13 February 2025).

2. **Which types of sources used here (existing software, research, or industry guidance) would be most relevant for your own project, and why?**

Suggested answers below

1. What types of sources have the group used in this section?
   - Existing commercial software tools (Upper, Routific).
   - Academic research (an IEEE journal article on vehicle routing with real-time traffic).
   - Industry and professional guidance on ethics, data privacy, and AI use.
2. How does the writer use these sources to justify features or decisions in their own project?
   - Existing apps are used to identify useful features the project will adopt or adapt.
   - Academic research is used to justify the inclusion of real-time traffic data.
   - Industry sources are used to inform ethical guidelines rather than just listing them.
3. What phrases or sentence structures show that the sources are being used to support the project, rather than just described?

   Some examples include…

   ✓ "We would take a similar approach by…"

   ✓ "which is a function we would want to implement…"

   ✓ "Research has shown that…"

   ✓ "To help inform the Ethical Guidelines, we reviewed…"

Task 4: Development & Implementation Summary

1. **Below are five functions of the Development & Implementation section (A – E), based on the Canvas guidance. Read the functions, then read sentences 1–10 and match each sentence to the function it demonstrates. Which words or phrases helped you decide? An example has been given.**

Functions of this section

A Describing system components
B Explaining how components are linked
C Stating technologies or tools
D Justifying design or technology choices
E Describing development workflow

Sentences from a group project

1) The project will be built with HTML, CSS and JavaScript.
   C Stating technologies or tools
2) A dynamically sized grid-based layout, and custom cell formatting with text entry into each cell.
3) Git will be ideal for efficient collaboration allowing our team to easily track updates to the code.
4) The UI will interact with the backend to perform complex formulae, file input and output, sorting, filtering and searching.
5) The development process follows a structured approach to ensure consistency, clarity and functionality.
6) Python will be used to manage the database written in SQL and to make calls to the Google Maps API.
7) The project is divided into the following sections: Physics engine, Graphical User Interface (GUI), Rendering Engine, and Saving and Data System.
8) The next stage of implementation will be to create an interface which provides the user with an easy-to-use interface.
9) Dear ImGUI library will be used for its lightweight, real-time capabilities, ensuring a smooth user experience.
10) The physics engine will be developed concurrently with a collision system that dynamically adjusts depending on object mass, velocity and trajectories.

Suggested answers below

1) The project will be built with HTML, CSS and JavaScript.
   C Stating technologies or tools
2) A dynamically sized grid-based layout, and custom cell formatting with text entry into each cell.
   A Describing system components
3) Git will be ideal for efficient collaboration allowing our team to easily track updates to the code.
   D Justifying design or technology choices
4) The UI will interact with the backend to perform complex formulae, file input and output, sorting, filtering and searching.
   B Explaining how components are linked
5) The development process follows a structured approach to ensure consistency, clarity and functionality.
   E Describing development workflow
6) Python will be used to manage the database written in SQL and to make calls to the Google Maps API.
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   A Describing system components
8) The next stage of implementation will be to create an interface which provides the user with an easy-to-use interface.
   E Describing development workflow
9) Dear ImGUI library will be used for its lightweight, real-time capabilities, ensuring a smooth user experience.
   D Justifying design or technology choices
10) The physics engine will be developed concurrently with a collision system that dynamically adjusts depending on object mass, velocity and trajectories.
   B Explaining how components are linked

Task 5: Data Sources

1. **Read sentences 1–8, which are taken from different group projects, and decide what each sentence is doing (a – d) in the Data Sources section. You can use more than one letter if**
   a) Identifies what data the system uses
   b) Explains where the data comes from
   c) Confirms permission or licensing
   d) Explains how confidentiality is ensured

Examples from student projects

1) All of our data is user-generated.
2) Passwords retrieved from the sign-up process will be encrypted and not stored in plaintext to ensure security.
3) Most of the data sources we will use are publicly available datasets, and we will generate data within the system to simulate various space interactions.
4) If user data is collected for testing or feedback purposes, it will be handled in compliance with GDPR or relevant data protection standards.
5) The system will use publicly available health datasets released under the Apache License 2.0, which permits their use for academic research and system development.
6) To protect privacy, the system will only display aggregated results, ensuring that no individual-level data can be identified.
7) User-generated data will include registration information and user activity within the platform, which will be collected through interactions with the system interface.
8) All personal data will be used solely for core system functionality and will not be disclosed publicly, with appropriate security measures in place to protect user information.

2. **Which words or phrases show that the group is planning how data will be handled later in the project, rather than just describing something already built?**

Suggested answers below

1. **Read sentences 1–8, which are taken from different group projects, and decide what each sentence is doing (a – d) in the Data Sources section.**

1. All of our data is user-generated.

   a) Identifies what data the system uses (maybe b as well…?)

2. Passwords retrieved from the sign-up process will be encrypted and not stored in plaintext to ensure security.

   d) Explains how confidentiality is ensured

3. Most of the data sources we will use are publicly available datasets, and we will generate data within the system to simulate various space interactions.

   a) Identifies what data the system uses

   b) Explains where the data comes from

4. If user data is collected for testing or feedback purposes, it will be handled in compliance with GDPR or relevant data protection standards.

   d) Explains how confidentiality is ensured

5. The system will use publicly available health datasets released under the Apache License 2.0, which permits their use for academic research and system development.

   b) Explains where the data comes from

   c) Confirms permission or licensing

6. To protect privacy, the system will only display aggregated results, ensuring that no individual-level data can be identified.

   d) Explains how confidentiality is ensured

7. User-generated data will include registration information and user activity within the platform, which will be collected through interactions with the system interface.

   a) Identifies what data the system uses

   b) Explains where the data comes from

8. All personal data will be used solely for core system functionality and will not be disclosed publicly, with appropriate security measures in place to protect user information.

   d) Explains how confidentiality is ensured

2. **Which words or phrases show that the group is planning how data will be handled later in the project?** [example of details]

Task 6: Testing and evaluation

1. **Below are real tutor comments from the Testing & Evaluation section of COMP208 projects. None of the projects received an A for this section. Read the comments and discuss:**
   - What are the most common problems identified by the tutors?
   - What do these comments suggest students should include in this section?

   a) "Consider adding in performance and memory testing for extra large spreadsheets to determine the limits of your application."

   b) "Testing section very generic. How about performance testing or any evaluation? Will you use beta testers and an evaluation survey?"

   c) "Mostly generic."

   d) "This is a simple list of testing strategies and, as a result, is way too generic."

   e) "A bit too vague."

   f) "This section lacks detail and it is very generic. It could apply to any project. Moreover, there is no mention of evaluation through beta testers."

Suggested answers below

Suggested answers below

What are the most common problems identified by the tutors?

- The testing sections are too generic and could apply to almost any project.
- Students often list testing types without explaining how they relate to their system.
- There is little or no evaluation, especially involving beta testers or user feedback.
- The section lacks project-specific detail (e.g. performance limits, realistic test cases).
- Testing is mentioned, but not clearly linked to system features or objectives.

What do these comments suggest students should include in this section?

- Project-specific testing, not just general testing terminology.
- Clear mention of performance testing, especially where system limits matter.
- An evaluation plan, including:
- beta testers,
- questionnaires or feedback methods.
- Brief explanations of what will be tested and why, rather than just naming test types.
- Language that shows planned testing and evaluation, not vague descriptions.

2. **Below are four sentences from Testing & Evaluation sections. Identify which two sentences are stronger, which two are weaker, and why. Choose one weaker sentence and suggest how to improvement it.**
   A. Model performance evaluation: Accuracy; Precision & Recall; AUC-ROC Score.
   B. This project requires rigorous testing to ensure accuracy, reliability and efficiency.
   C. Unit tests will be run during development to validate core Python and JavaScript components before integration.
   D. We will create a questionnaire covering features, performance, and accessibility, which external testers will complete after using the system. The responses will be used to evaluate whether the system meets relevant project objectives related to usability and performance.

Suggested answers below

Suggested answers below

A. Model performance evaluation: Accuracy; Precision & Recall; AUC-ROC Score.
   Weaker
   Just a list of metrics with no explanation of how, when, or why they will be used. It could apply to many projects and lacks context.
B. This project requires rigorous testing to ensure accuracy, reliability and efficiency.
   Weaker
   Very generic and vague. It states that testing is important but does not explain what kind of testing will be carried out or how it relates to the specific system.
C. Unit tests will be run during development to validate core Python and JavaScript components before integration.
   Stronger
   Explains when testing will happen (during development), what will be tested (core Python and JavaScript components), and why (before integration). It is specific and project-focused.
D. We will create a questionnaire covering features, performance, and accessibility, which external testers will complete after using the system. The responses will be used to evaluate whether the system meets relevant project objectives related to usability and performance.
   Stronger
   Clearly describes an evaluation method (a questionnaire), who will be involved (external testers), and how the results will be used (to evaluate whether objectives are met). This directly addresses tutor feedback about evaluation and beta testing.

Example improvement of sentence B

Original: “This project requires rigorous testing to ensure accuracy, reliability and efficiency.”

Improved version:
“Unit and integration testing will be carried out during development to ensure the accuracy and reliability of the system’s core features, including data processing and user interactions.”

Final tips for the other sections

Below are tips for the sections we have not covered in today's session.

Ethical Considerations

- Focus on ethical issues **specific to your project**, not generic ethics.
- Use clear obligation language (*will comply with*, *will ensure*).

BCS Project Criteria

- Link each criterion to **concrete features or methods** in your project.
- Use evaluative language (*demonstrates*, *supports*), not vague claims.

UI / UX Mockup

- Explain **what the interface shows and how users interact with it**.
- Label and refer to mockups clearly in the text.

Project Plan & Risks

- Describe tasks and dependencies **clearly and specifically**.
- Write risks as **if–then** statements with clear contingencies.